



Mount Kenya

University

Denis Kibichiy

BSCCS/2024/55349

BSC Computer Science

Unit Code: BIT3208 Advanced Web Design and Development

Michael Nyoro

May/August 2025/2026

Project Details

Project Title: Runi Empire Grooming Barbershop Management System

Selected Technology: JavaScript, PHP, MySQL, XAMPP

GitHub Link:

https://github.com/deniskibichiy/BIT3208_Advanced_Web_Design_and_Development#

Table of Contents

1. Week 1: Environment setup and database connection.....	4
1.1. Figure 1: XAMPP Installation and Setup with MySQL in Linux.....	4
1.2. Figure 2: Apache + MySQL Running and Operational.....	4
1.3. Figure 3: Index.php successfully rendered.....	5
1.4. Figure 4: Database connection successful.....	5
1.5. Figure 5: Week1 database.....	6
Week 1 reflection:.....	6
2. Week 2: User Interface Planning and System Design.....	7
2.1. Figure 2.1: Homepage User Interface Mockup.....	7
2.2. Figure 2.2: Login Page User Interface Mockup.....	8
2.3. Figure 2.3: Admin Dashboard User Interface Mockup.....	9
2.4. Figure 2.4: Homepage Figma prototype.....	10
2.5. Figure 2.5: Login page Figma prototype.....	11
2.6. Figure 2.6: Dashboard Figma prototype.....	12
2.7. Figure 2.7: Color Scheme.....	13
2.8. Figure 2.8: Navigation Structure.....	13
Reflection week 2:.....	13
3. Week 3: JavaScript and PHP Basics.....	15
3.1. Figure 1: Invalid Login Validation.....	15
3.2. Fig 2: Valid login input accepted.....	15
3.3. Fig 3: DOM manipulation interaction.....	16
3.5. Figure 5: Week 3 database connection success evidence.....	16
Week 3 Reflection:.....	17
4. WEEK 4: Server-Side Components and Backend Development Foundations.....	18
4.1. Figure 1: Login page initial state.....	18
4.2. Figure 2: Failed Login Attempt.....	18
4.3. Figure 3: Successful login attempt - Redirected to dashboard.....	18
4.4. Figure 4: Session-based admin dashboard.....	19
4.5. Figure 5: Folder Structure.....	19
Week 4 Reflection:.....	19
5. Week 5: Database Components and CRUD Operations.....	20
5.1. Figure 5.1: Databases Screenshot.....	20
5.2. Figure 5.2: Created Tables within Database and defined their schema.....	20
5.3. Figure 5.3: CRUD Operations on database.....	21
5.4. Figure 5.4: Database Connection.....	21
5.5. Figure 5.5: Successful login through the database.....	22

Week 5 Reflection:.....	22
6. Week 6: Database Integration and CRUD Operations.....	23
6.1. Database Creation (Already done in week 5 but now normalized).....	23
6.2. Entity Relationship Diagram.....	24
6.3. User Registration Form on Submit submits to the Users table.....	24
6.4. Admin can display users on the view users page: NOTE: Last user.....	25
6.5. Update and Delete Records via the terminal evidence.....	25
Week 6 reflection:.....	26

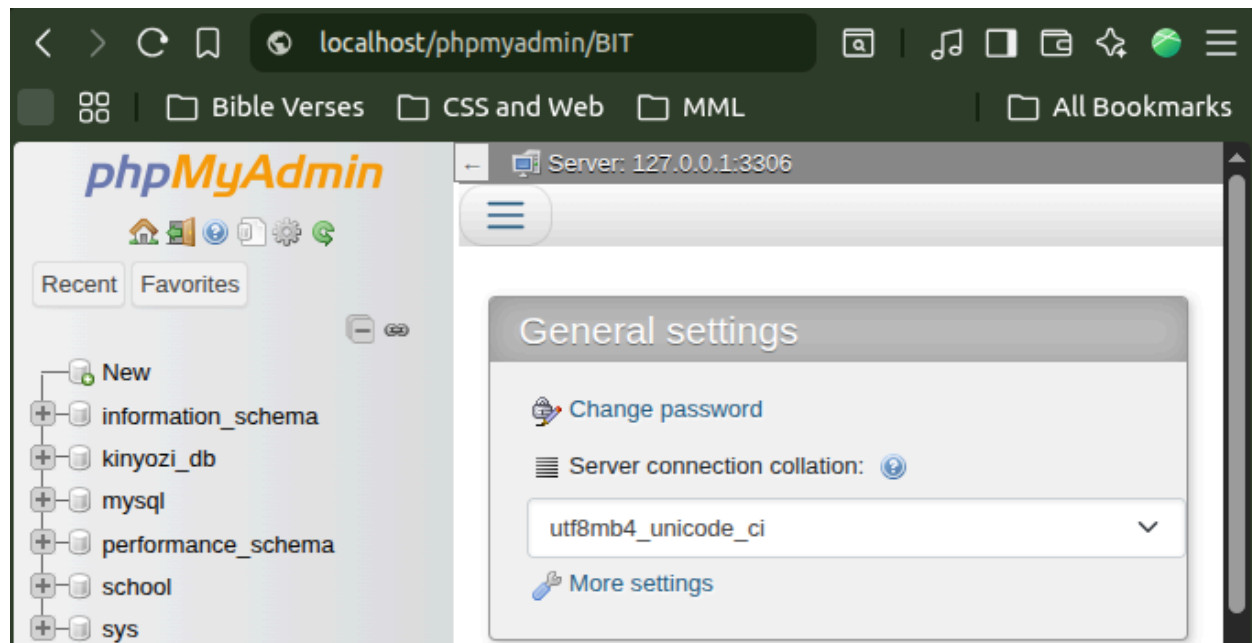
1. Week 1: Environment setup and database connection

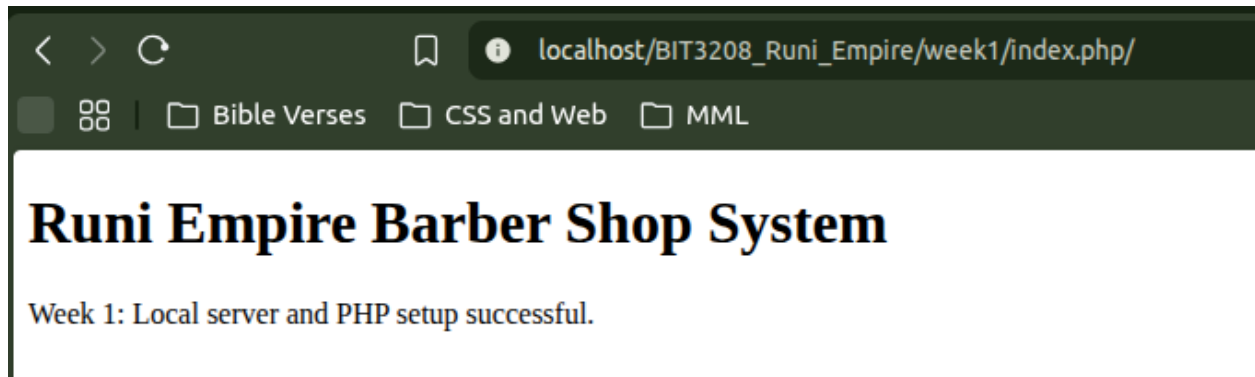
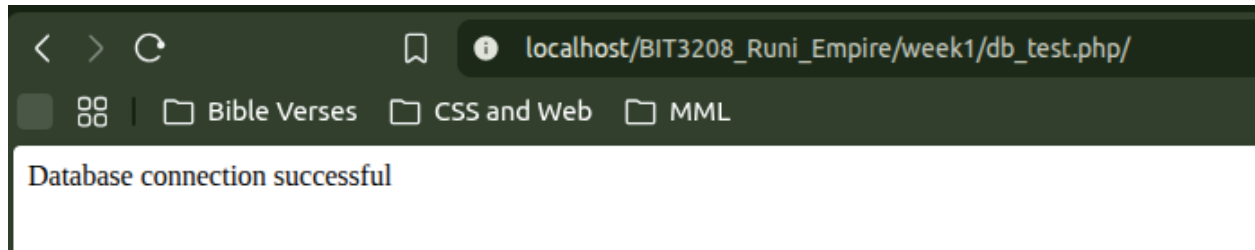
1.1. Figure 1: XAMPP Installation and Setup with MySQL in Linux

```
denis@denis-HP-ProBook-450-G7:~$ sudo /opt/lampp/xampp status
Version: XAMPP for Linux 8.2.12-0
Apache is running.
MySQL is not running.
ProFTPD is not running.
denis@denis-HP-ProBook-450-G7:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/lib/systemd/system/mysql.service; enabled; vendor p
   Active: active (running) since Thu 2026-05-28 13:18:55 EAT; 39min ago
   Main PID: 10633 (mysqld)
   Status: "Server is operational"
   Tasks: 39 (limit: 18907)
   Memory: 434.2M
   CPU: 14.863s
   CGroup: /system.slice/mysql.service
           └─10633 /usr/sbin/mysqld

May 28 13:18:53 denis-HP-ProBook-450-G7 systemd[1]: Starting MySQL Commu
May 28 13:18:55 denis-HP-ProBook-450-G7 systemd[1]: Started MySQL Commu
denis@denis-HP-ProBook-450-G7:~$ ss -tulnp | grep 3306
tcp  LISTEN 0      70          127.0.0.1:3306  0.0.0.0:*
tcp  LISTEN 0      151         127.0.0.1:3306  0.0.0.0:*
denis@denis-HP-ProBook-450-G7:~$
```

1.2. Figure 2: Apache + MySQL Running and Operational



1.3. Figure 3: Index.php successfully rendered**1.4. Figure 4: Database connection successful**

1.5.Figure 5: Week1 database

```
mysql> USE week1db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_week1db |
+-----+
| users              |
+-----+
1 row in set (0.01 sec)

mysql> DESCRIBE users
-> ;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id    | int  | NO   | PRI | NULL    | auto_increment |
| name  | varchar(100) | YES |     | NULL    |
| phone_number | varchar(20) | YES |     | NULL    |
| created_at | timestamp | YES |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

Week 1 reflection:

During week one, the focus was on setting up the environment to facilitate practical activities which are part of the Unit requirements. A key challenge I experienced is that the XAMPP MySQL couldn't load successfully due to port clashes with the system MySQL. To solve this, I had to configure Apache to use system MySQL instead of the one provided by PHPMyAdmin, and the process was successful.

2. Week 2: User Interface Planning and System Design

2.1. Figure 2.1: Homepage User Interface Mockup



2.2.Figure 2.2: Login Page User Interface Mockup



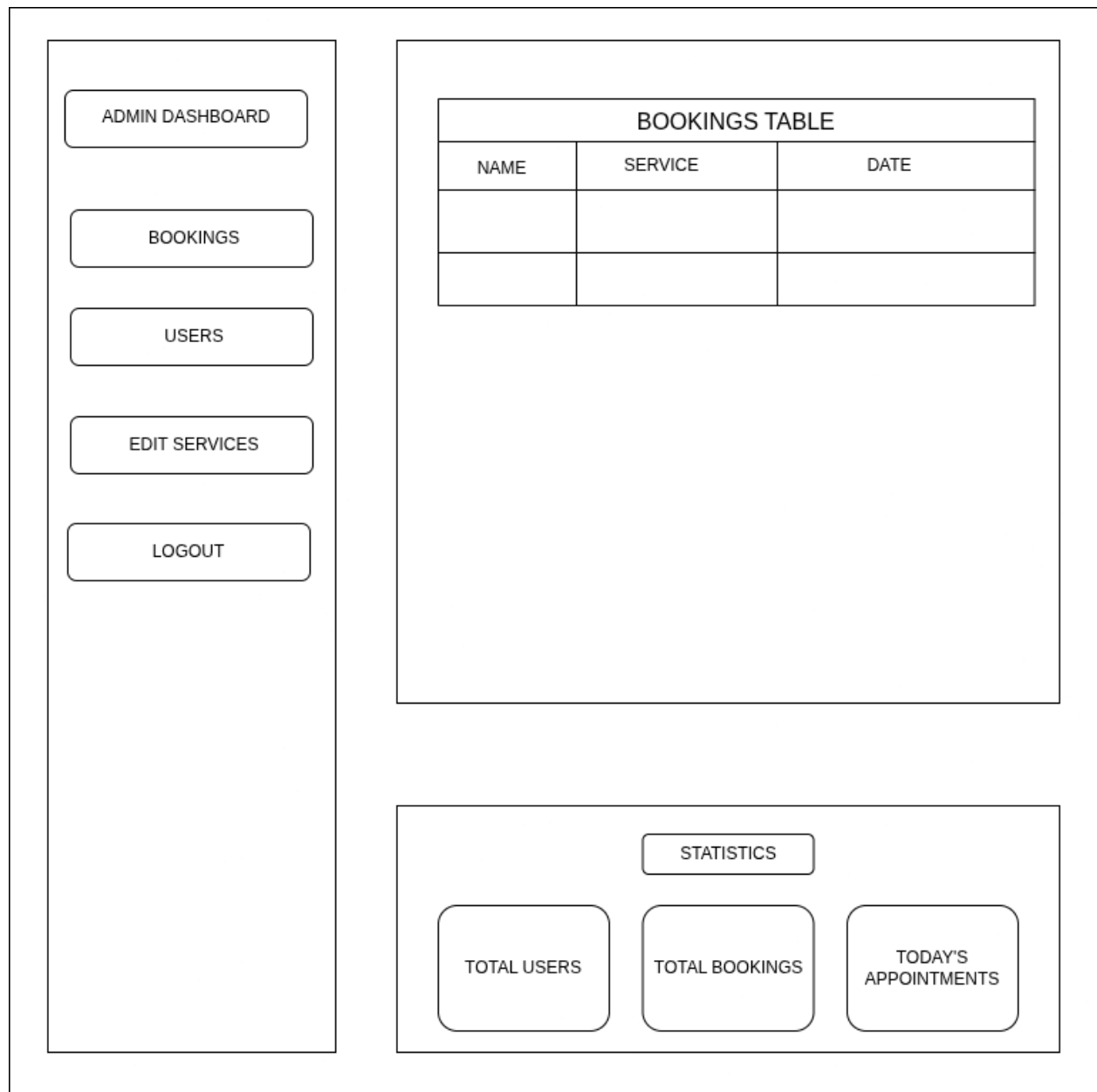
A user interface mockup for a login page. It features a central container with a black border. Inside, the text "WELCOME TO RUNI EMPIREE" is centered at the top. Below this, the label "Username:" is followed by a rectangular input field. The label "Password:" is followed by another rectangular input field. At the bottom, there is a wide rectangular button with the text "LOGIN" centered on it.

WELCOME TO RUNI EMPIREE

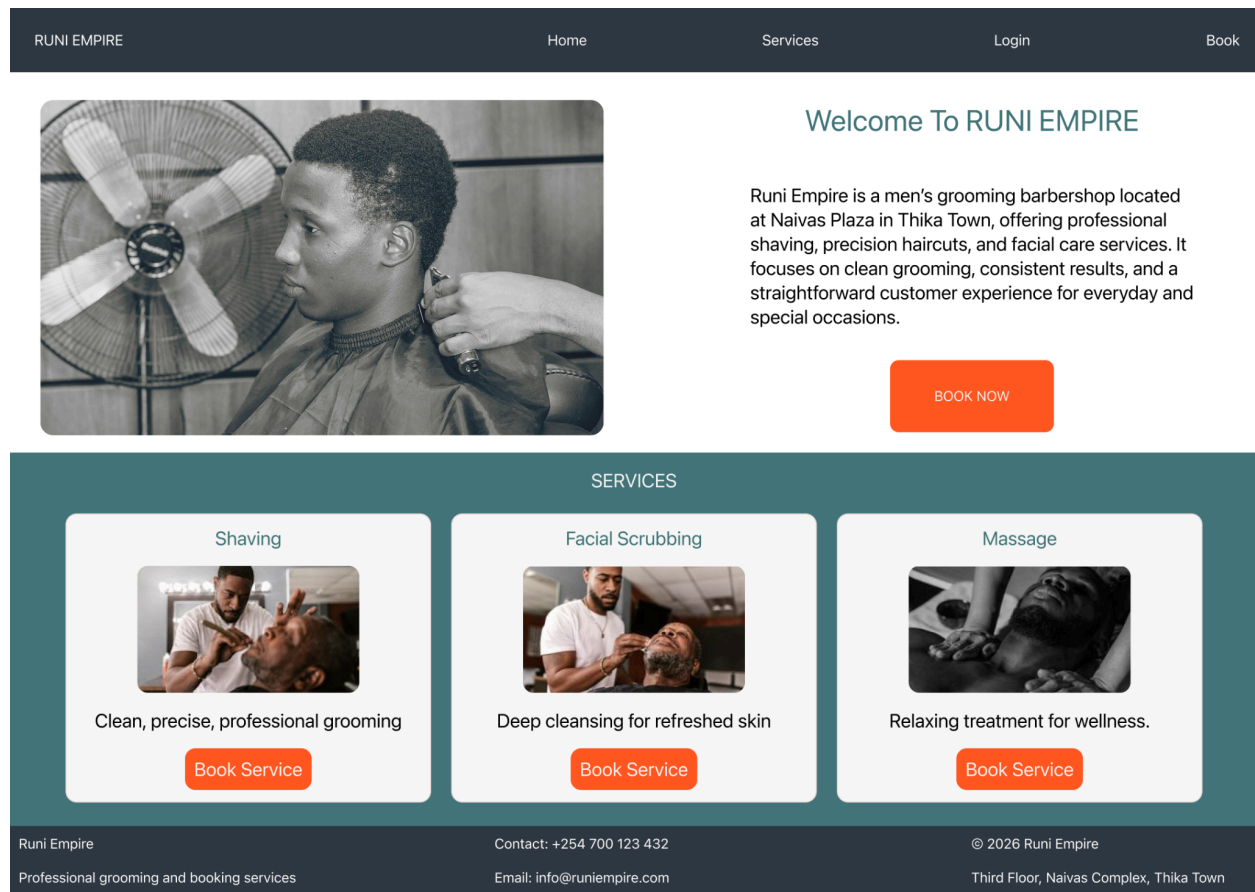
Username:

Password:

LOGIN

2.3. Figure 2.3: Admin Dashboard User Interface Mockup

2.4. Figure 2.4: Homepage Figma prototype



2.5. Figure 2.5: Login page Figma prototype

RUNI EMPIRE

Home

Services

Login

Book

Login to Access Our Services

Email:

Password:

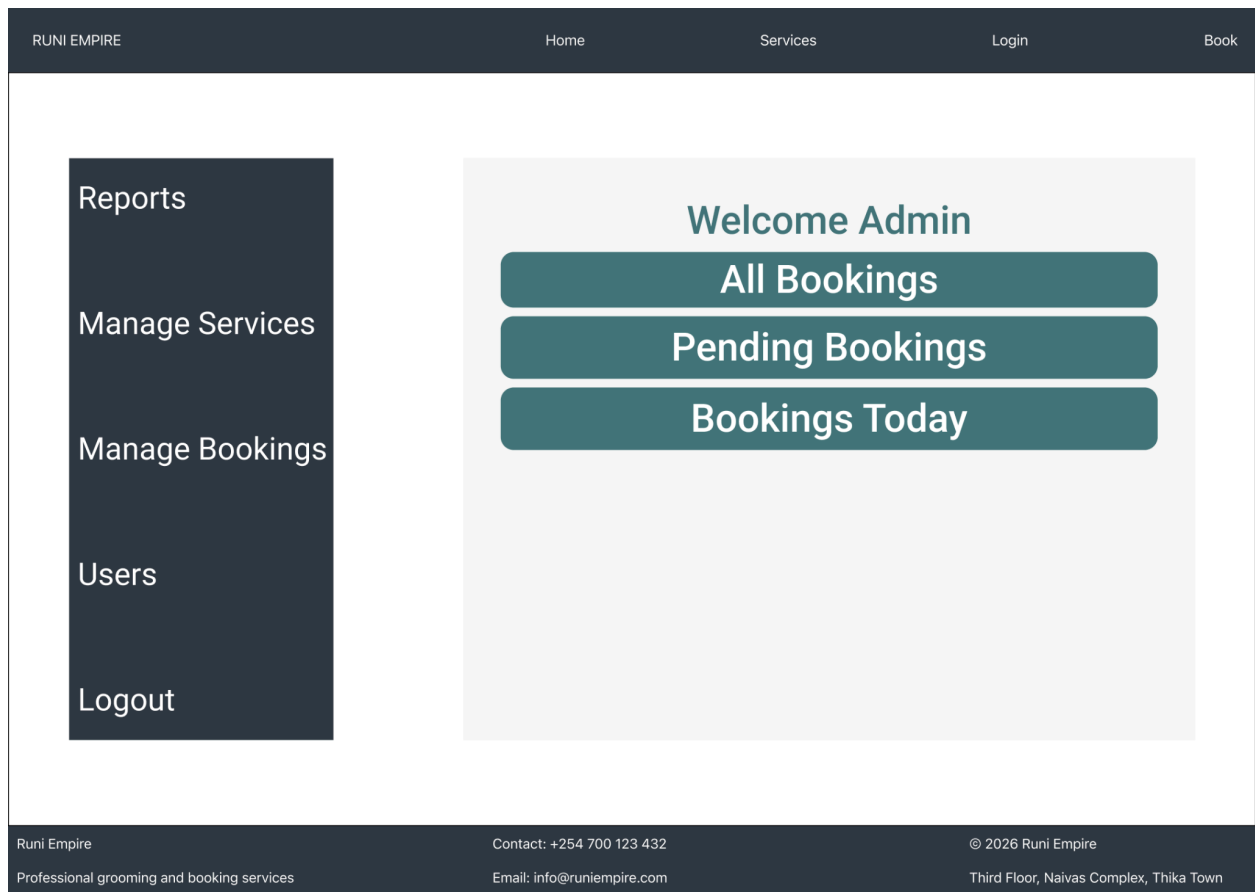
Login

Don't have an account? [Sign Up](#)

Runi Empire
Professional grooming and booking services

Contact: +254 700 123 432
Email: info@runiempire.com

© 2026 Runi Empire
Third Floor, Naivas Complex, Thika Town

2.6. Figure 2.6: Dashboard Figma prototype

2.7. Figure 2.7: Color Scheme

- i. Background: #F5F5F5 : Light Gray
- ii. Secondary/Highlights: #76ABAE: Teal
- iii. Main text: #303841: Charcoal
- iv. Accent/Actions: #FF5722: Orange

2.8. Figure 2.8: Navigation Structure

Reflection week 2:

Week 2 provided me with another key challenge in that after visualizing how my website should be laid out, the next obstacle was designing the prototype using Figma. To complete the process successfully, I had to consider carefully how to come up with reusable components to avoid creating frames repetitively.

3. Week 3: JavaScript and PHP Basics

3.1. Figure 1: Invalid Login Validation

Create Account to Access our Services

Name	Denis Kibichiy	
Email	denismutai5@gmail.com	
Phone Number	+254708198361	
Password	*****	
Confirm Password	*****	Passwords must match

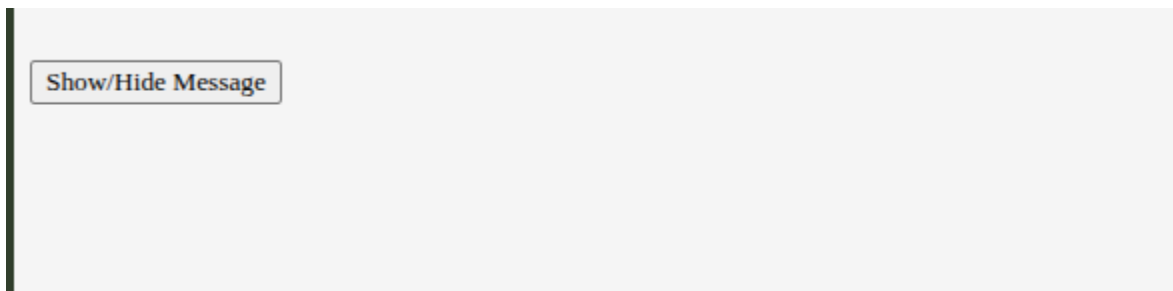
Already have an account? [Login](#)

3.2. Fig 2: Valid login input accepted

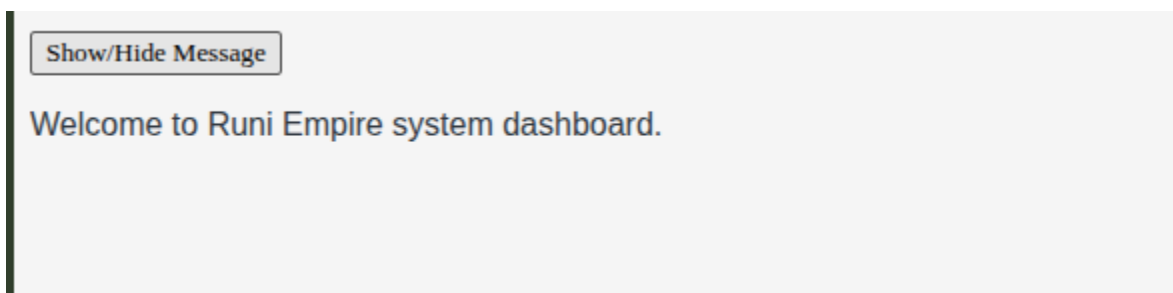
Create Account to Access our Services

Name	Denis Kibichiy	
Email	denismutai5@gmail.com	
Phone Number	+254708198361	
Password	*****	
Confirm Password	*****	

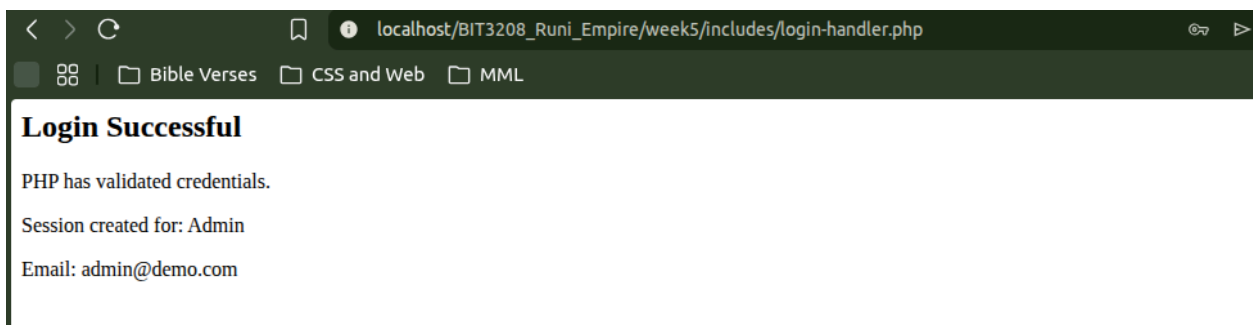
3.3.Fig 3: DOM manipulation interaction



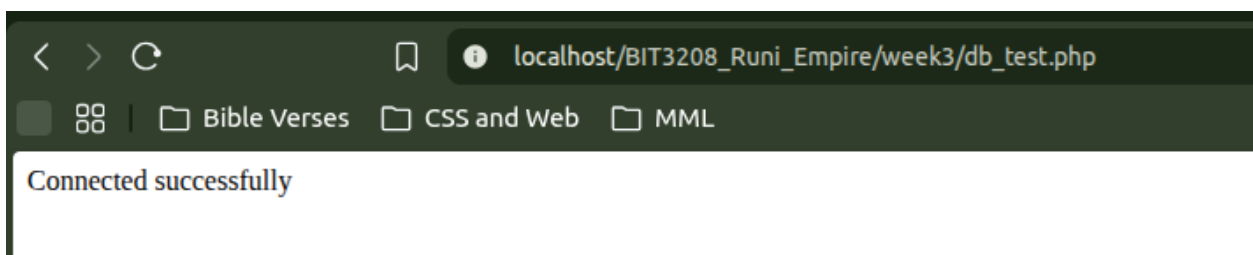
After button click



3.4. Figure 4: PHP Interaction Evidence after valid login details captured



3.5.Figure 5: Week 3 database connection success evidence

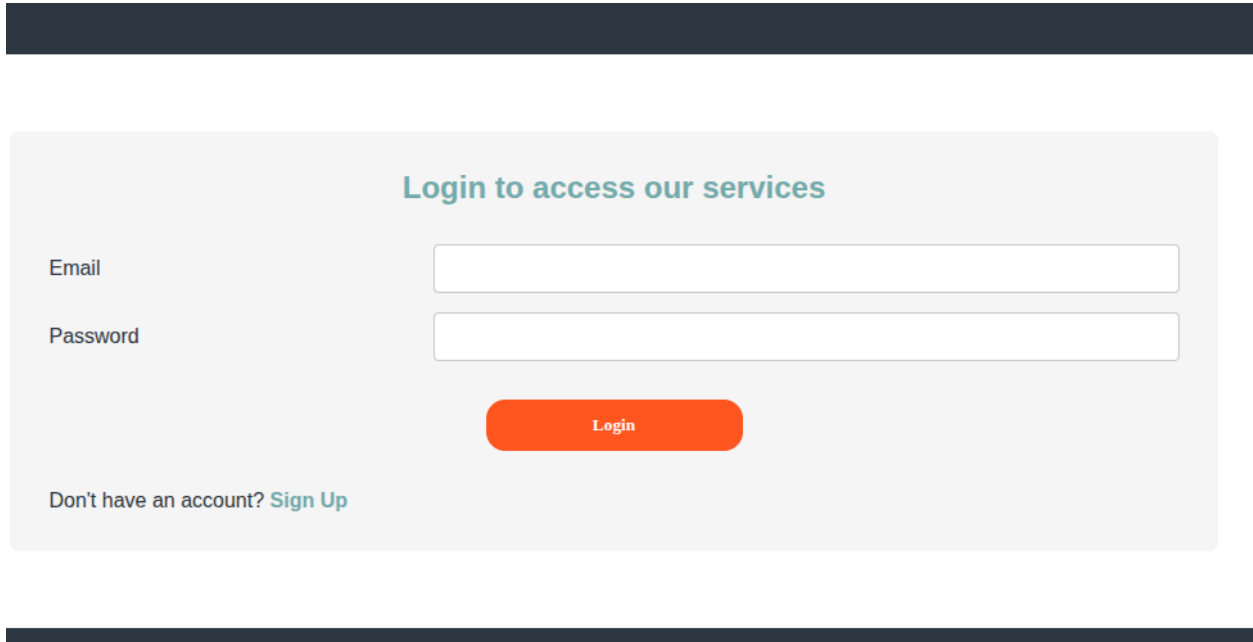


Week 3 Reflection:

In week 3, much of the focus shifted to JavaScript and PHP processing. The most challenging aspect was using JavaScript to perform form validation, and to succeed, I had to learn a lot of concepts, mostly dealing with DOM Manipulation such as query selectors, creating nodes, and appending them to already existing parent containers.

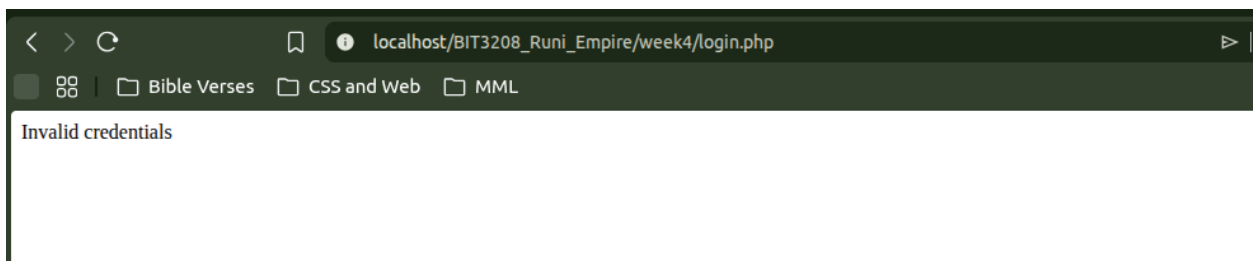
4. WEEK 4: Server-Side Components and Backend Development Foundations

4.1. Figure 1: Login page initial state

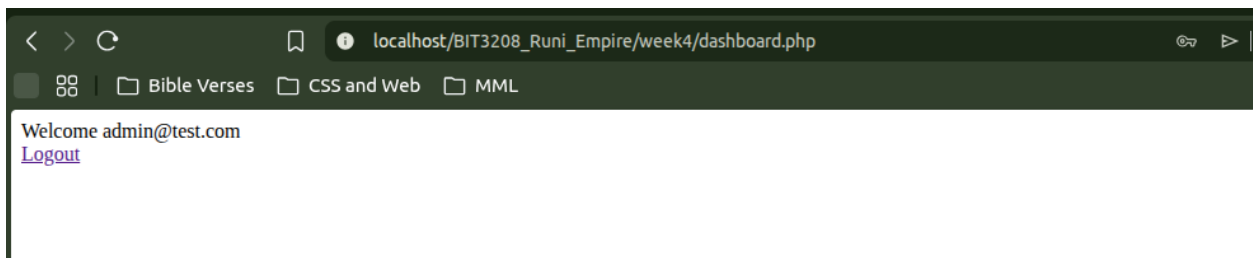


The image shows a login page with a dark blue header. Below the header is a light gray box containing the login form. The form has the title "Login to access our services" in teal. It includes two input fields: "Email" and "Password". Below the "Password" field is an orange "Login" button. At the bottom of the form, there is a link: "Don't have an account? [Sign Up](#)".

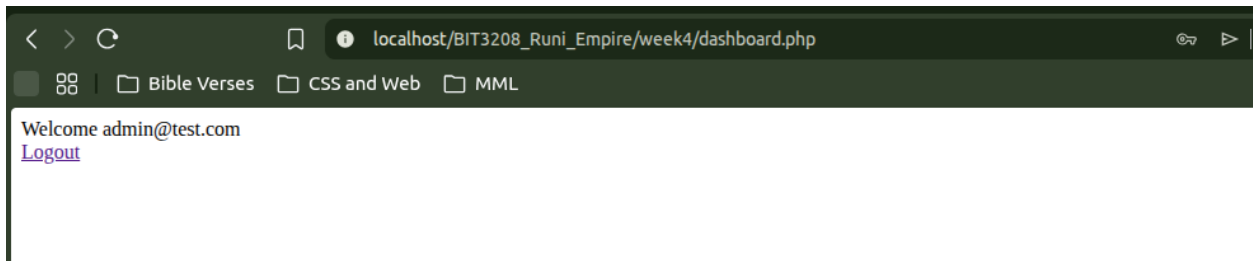
4.2. Figure 2: Failed Login Attempt



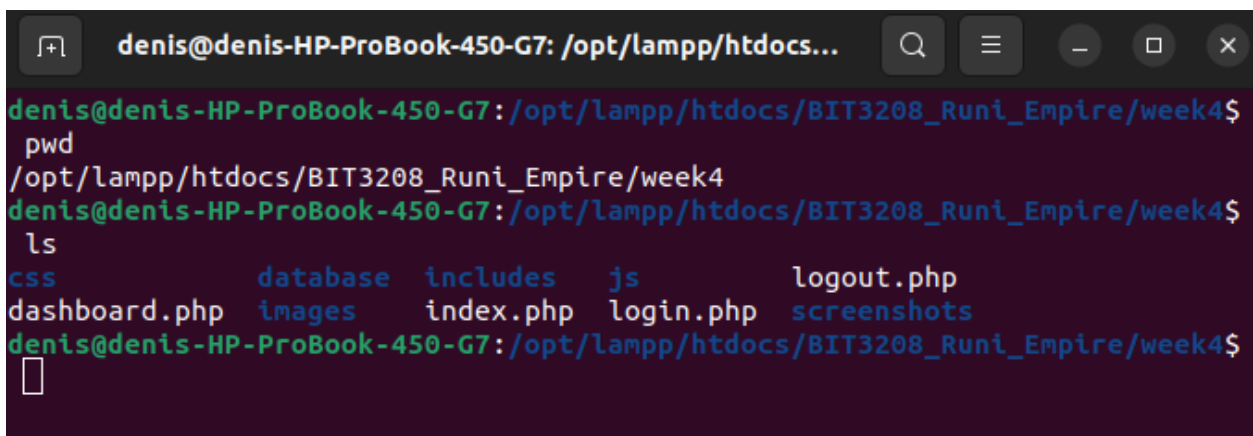
4.3. Figure 3: Successful login attempt - Redirected to dashboard



4.4. Figure 4: Session-based admin dashboard



4.5. Figure 5: Folder Structure



Week 4 Reflection:

In Week 4, the focus was on form processing, authentication basics, and sessions in PHP. The most difficult part was understanding how data moves from HTML forms into PHP and how validation is applied before processing. I learned how to capture form input using POST, validate it, and manage user sessions to maintain login state. I also built a basic login system and session-protected dashboard. This helped me understand how authentication systems control access to different parts of a web application.

5. Week 5: Database Components and CRUD Operations

5.1. Figure 5.1: Databases Screenshot

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| kinyozi_db |
| mysql |
| performance_schema |
| runi_empire_db |
| school |
| sys |
| week1db |
| week_one |
+-----+
9 rows in set (0.01 sec)
```

5.2. Figure 5.2: Created Tables within Database and defined their schema

```
mysql> USE runi_empire_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_runi_empire_db |
+-----+
| bookings |
| users |
+-----+
2 rows in set (0.00 sec)

mysql> DESCRIBE users;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id | int | NO | PRI | NULL | auto_increment |
| first_name | varchar(100) | YES | | NULL | |
| last_name | varchar(100) | YES | | NULL | |
| email | varchar(100) | YES | UNI | NULL | |
| phone_number | varchar(20) | YES | | NULL | |
| password | varchar(255) | YES | | NULL | |
| created_at | timestamp | YES | | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)
```

5.3. Figure 5.3: CRUD Operations on database

```
mysql> INSERT INTO users (first_name, last_name, email, phone_number, password)
-> VALUES ('Brian', 'Kiptoo', 'brian.kiptoo@gmail.com', '0712345678', '1234');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM users;
+-----+-----+-----+-----+-----+-----+-----+
| id | first_name | last_name | email | phone_number | password | created_at |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | Brian | Kiptoo | brian.kiptoo@gmail.com | 0712345678 | 1234 | 2026-06-05 11:47:07 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> UPDATE users
-> SET first_name = 'Kevin',
-> last_name = 'Mutua',
-> WHERE id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

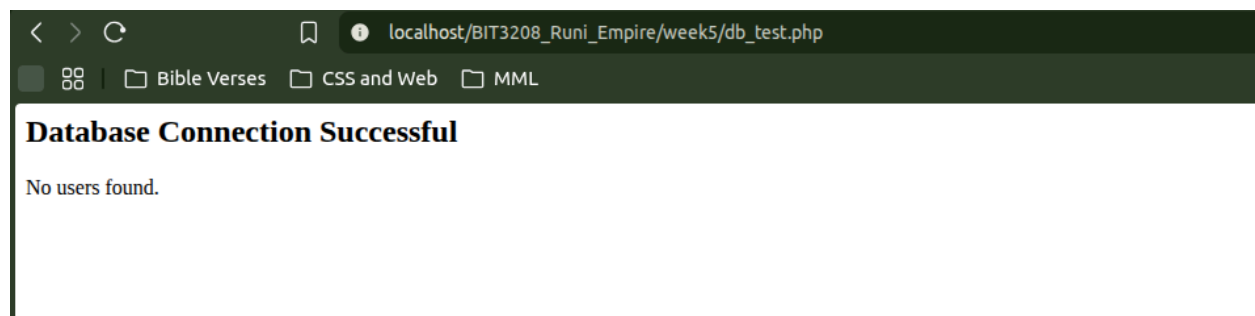
mysql> SELECT * FROM users;
+-----+-----+-----+-----+-----+-----+-----+
| id | first_name | last_name | email | phone_number | password | created_at |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | Kevin | Mutua | brian.kiptoo@gmail.com | 0712345678 | 1234 | 2026-06-05 11:47:07 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

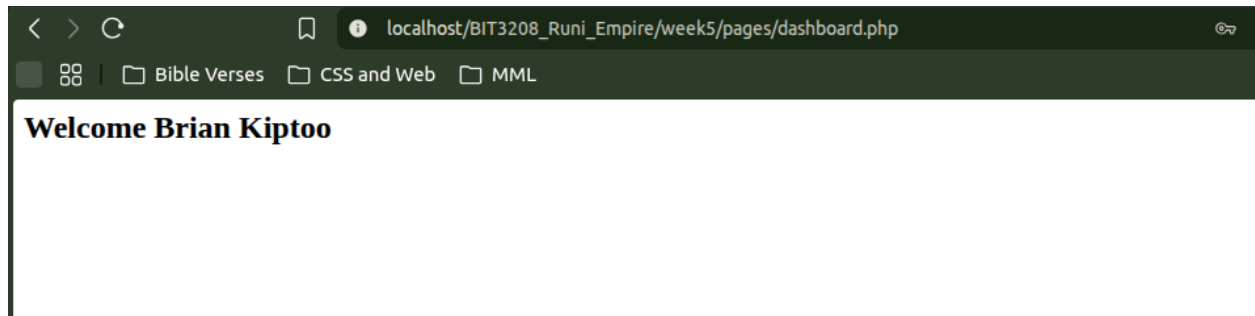
mysql> DELETE FROM users
-> WHERE id = 1;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM users;
Empty set (0.00 sec)

mysql>
```

5.4. Figure 5.4: Database Connection



5.5. Figure 5.5: Successful login through the database**Week 5 Reflection:**

In Week 5, I focused on databases and integrating them with PHP applications. The main challenge was understanding SQL operations and how PHP interacts with MySQL databases. I learned how to create databases and tables, perform CRUD operations, and connect PHP to a database using PDO. I also implemented database-driven login functionality and data retrieval. This improved my understanding of backend development and data management in web applications.

6. Week 6: Database Integration and CRUD Operations

6.1. Database Creation (Already done in week 5 but now normalized)

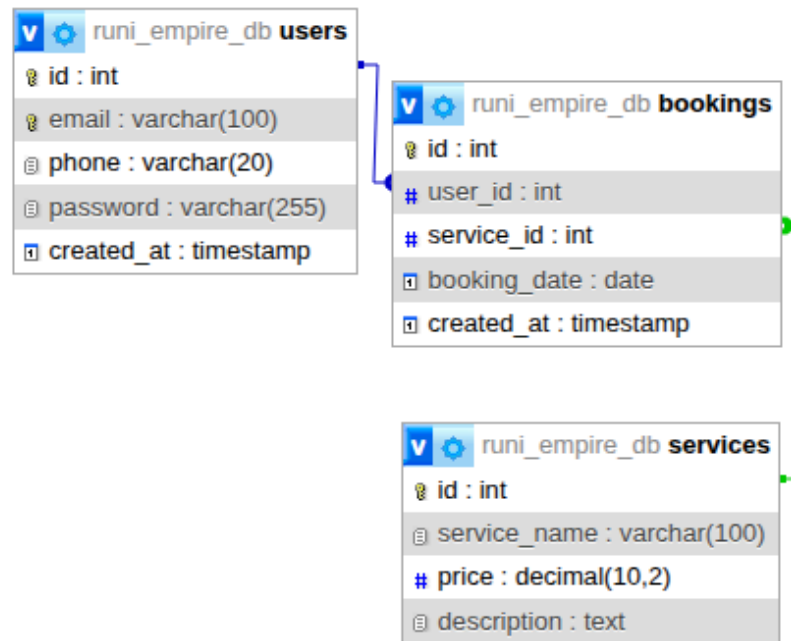
Users table normalized

```
mysql> DESCRIBE users;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default          | Extra          |
+-----+-----+-----+-----+-----+-----+
| id    | int           | NO   | PRI | NULL             | auto_increm   |
| email | varchar(100)  | YES  | UNI | NULL             |               |
| phone | varchar(20)   | YES  |     | NULL             |               |
| password | varchar(255) | YES  |     | NULL             |               |
| created_at | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.01 sec)
```

New Bookings Table Introduced Using FK

```
mysql> CREATE TABLE bookings (
  ->   id INT AUTO_INCREMENT PRIMARY KEY,
  ->   user_id INT NOT NULL,
  ->   service_id INT NOT NULL,
  ->   booking_date DATE NOT NULL,
  ->   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  ->
  ->   CONSTRAINT fk_booking_user
  ->     FOREIGN KEY (user_id)
  ->     REFERENCES users(id),
  ->
  ->   CONSTRAINT fk_booking_service
  ->     FOREIGN KEY (service_id)
  ->     REFERENCES services(id)
  -> );
Query OK, 0 rows affected (0.06 sec)
```

6.2. Entity Relationship Diagram



6.3. User Registration Form on Submit submits to the Users table

User Inputting the details

Create Account to Access our Services

Name

Georgina Thuo

Email

georginathuo@gmail.com

Phone Number

0712345678

Password

.....

Confirm Password

.....

Sign Up

User appearing on the table

```
mysql> SELECT * FROM users WHERE email = 'georginathuo@gmail.com';
+----+-----+-----+-----+-----+-----+
| id | name          | email                | phone      | password    |
|    |              |                     | created_at |             |
+----+-----+-----+-----+-----+-----+
|  7 | Georgina Thuo | georginathuo@gmail.com | 0712345678 | $2y$10$V871xLyLYPF.JHWBg9QDSeCPVLW91DY1MVwSwlksL5sMx6NVcmHzu | 2026-06-18 11:49:45 |
+----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

6.4. Update and Delete Records via the terminal evidence

```
mysql> UPDATE users
-> SET name = 'Brian Kiptoo'
-> WHERE email = 'brian.kiptoo@gmail.com';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE users
-> SET name = 'Faith Wanjiru'
-> WHERE email = 'faith.wanjiru@gmail.com';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT id, name, email, phone FROM users;
+----+-----+-----+-----+
| id | name          | email                | phone      |
+----+-----+-----+-----+
|  2 | Brian Kiptoo  | brian.kiptoo@gmail.com | 0712345678 |
|  3 | Faith Wanjiru | faith.wanjiru@gmail.com | 0722113344 |
|  4 | NULL         | kevin.mutua@gmail.com  | 0700123456 |
|  5 | NULL         | amina.mohamed@gmail.com | 0799887766 |
|  6 | NULL         | dennis.otieno@gmail.com | 0733557799 |
|  7 | Georgina Thuo | georginathuo@gmail.com | 0712345678 |
+----+-----+-----+-----+
6 rows in set (0.00 sec)
```

Delete

```
mysql> DELETE FROM users
      -> WHERE email != 'georginathuo@gmail.com';
Query OK, 5 rows affected (0.02 sec)

mysql> SELECT * FROM users;
+----+-----+-----+-----+-----+
| id | name          | email                  | phone      | password |
|    |              |                       | created_at |          |
+----+-----+-----+-----+-----+
|  7 | Georgina Thuo | georginathuo@gmail.com | 0712345678 | $2y$10$V871xL |
|    |              |                       |            | yLYPF.JHWBg9QDSeCPVLW91DY1MVwSwlksL5sMx6NVcmHzu |
|    |              |                       |            | 2026-06-18 11:49:45 |
+----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> █
```

Week 6 reflection:

In README

7. Week 7: User Authentication and Session Management

7.1. Figure 7.1: users table with different roles included

```
mysql> DESCRIBE users;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
name	varchar(100)	YES		NULL	
email	varchar(100)	YES	UNI	NULL	
phone	varchar(20)	YES		NULL	
password	varchar(255)	YES		NULL	
created_at	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED
role	enum('user','admin','super_admin')	YES		user	

```
7 rows in set (0.00 sec)
```

7.2. Figure 7.2: Role assignments done successfully

```
mysql> SELECT id, name, email, role FROM users;
```

id	name	email	role
1	Georgina Thuo	georginathuo@gmail.com	super_admin
2	Denis Kibichiy	denismutai5@gmail.com	admin

```
2 rows in set (0.00 sec)

mysql>
```

7.3. Figure7.3: Normal User Registration

Create Account to Access our Services

Name

Email

Phone Number

Password

Confirm Password

Already have an account? [Login](#)

7.4. Figure 7.4: Normal User saved with role successfully

```
mysql> SELECT id,name,email,role
-> FROM users;
+-----+-----+-----+-----+
| id | name          | email                | role      |
+-----+-----+-----+-----+
| 1 | Georgina Thuo | georginathuo@gmail.com | super_admin |
| 2 | Denis Kibichiy | denismutai5@gmail.com | admin      |
| 3 | Elvis Kipkosgei | elvis1234@gmail.com | user      |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

7.5. Figure 7.5: Password hashing for security purposes

```
mysql> SELECT email,password
-> FROM users;
+-----+-----+-----+-----+
| email                | password |
+-----+-----+-----+-----+
| georginathuo@gmail.com | $2y$10$BCKp5kUA47Lt4wPZlPizY04.0yA5SgyxCeYTr3.u4jpSYoz2NRXsu |
| denismutai5@gmail.com | $2y$10$XH9wa0zzp1U.1XzD80Xrm.xQQkD/7lVAy64aHtBccZWWhQZQsfTVS |
| elvis1234@gmail.com   | $2y$10$P7CWHU3V2Nr2FxFYrUSua4eFh3l9coEbyj/W2zmMMz25RC0oKbUyWe |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

7.6. Week 7 reflection

In README (root README)

7.7.